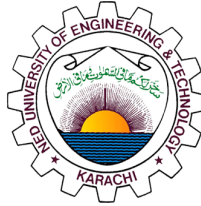


NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Information Technology



LAB REPORT: SESSION 07

Experimenting with Digital Multiplexer IC

Course Code: CS-251

Section: F

Lab Session: 7

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1 Multiplexers (MUX)

1.1 Theory and Operational Principles

A digital data Multiplexer (MUX) is a combinational circuit designed to select one of several data inputs and route it to a single output line. The selection process is governed by a set of data-select inputs. In a standard configuration, a multiplexer with 2^n input lines requires n selection lines to determine which input bit is connected to the output.

1.2 74150 - 16 x 1 Multiplexer

The 74150 IC is a 16-line-to-1-line multiplexer. It consists of sixteen data inputs and four data-selection lines. This IC is particularly useful for data routing and the implementation of complex Boolean functions.

Functional Pin Description:

- **E0 through E15:** Sixteen active-high data input lines.
- **A, B, C, D:** Four active-high data select lines (D is the MSB).
- **W:** Active-low output line.
- **G':** Active-low enable line (Strobe).
- **VCC/GND:** Standard supply and ground connections.

2 Electronics Workbench Diagrams

The logic behavior of the multiplexer circuits was simulated using Electronics Workbench. This software verification confirms that the data selection logic behaves according to the established truth tables.

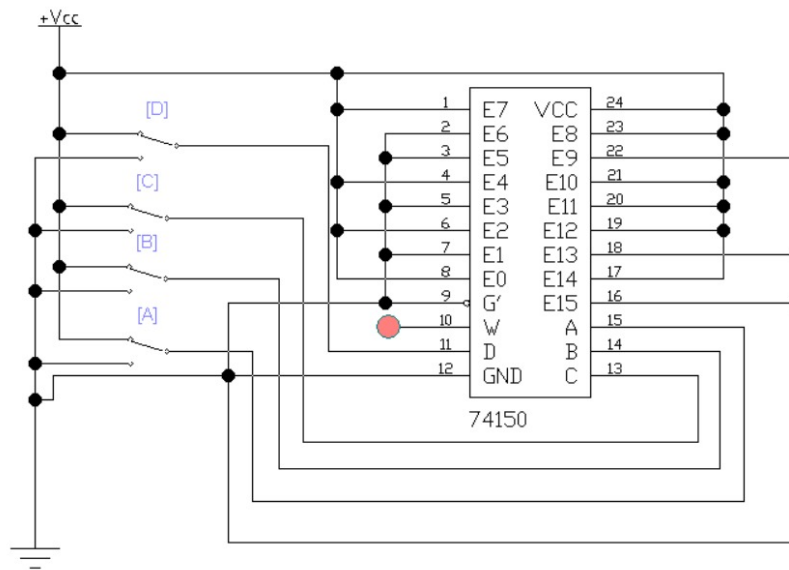


Figure 1: Electronics Workbench simulation of the 74150 Multiplexer circuit.

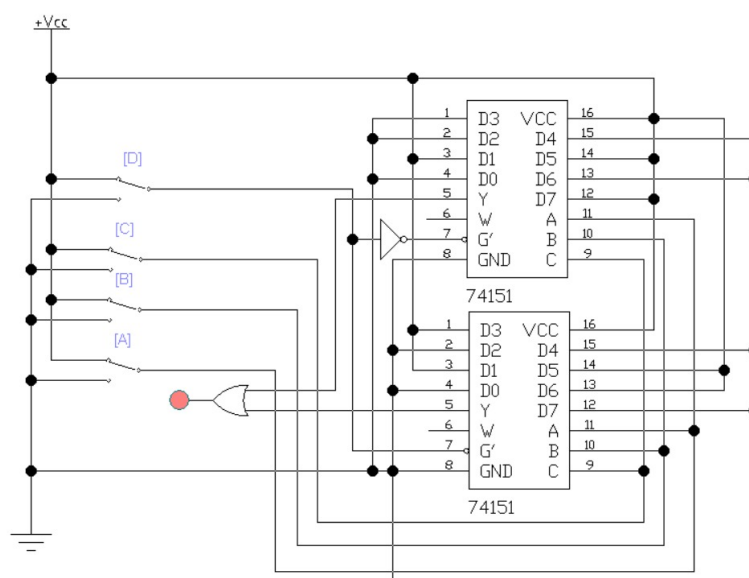


Figure 2: Simulation of logic function implementation using 74151 ICs.

3 MUX Hardware

The physical implementation of the multiplexer circuit was performed on a breadboard. This setup utilized the 74150/74151 IC series to verify the theoretical logic in a real-world environment.

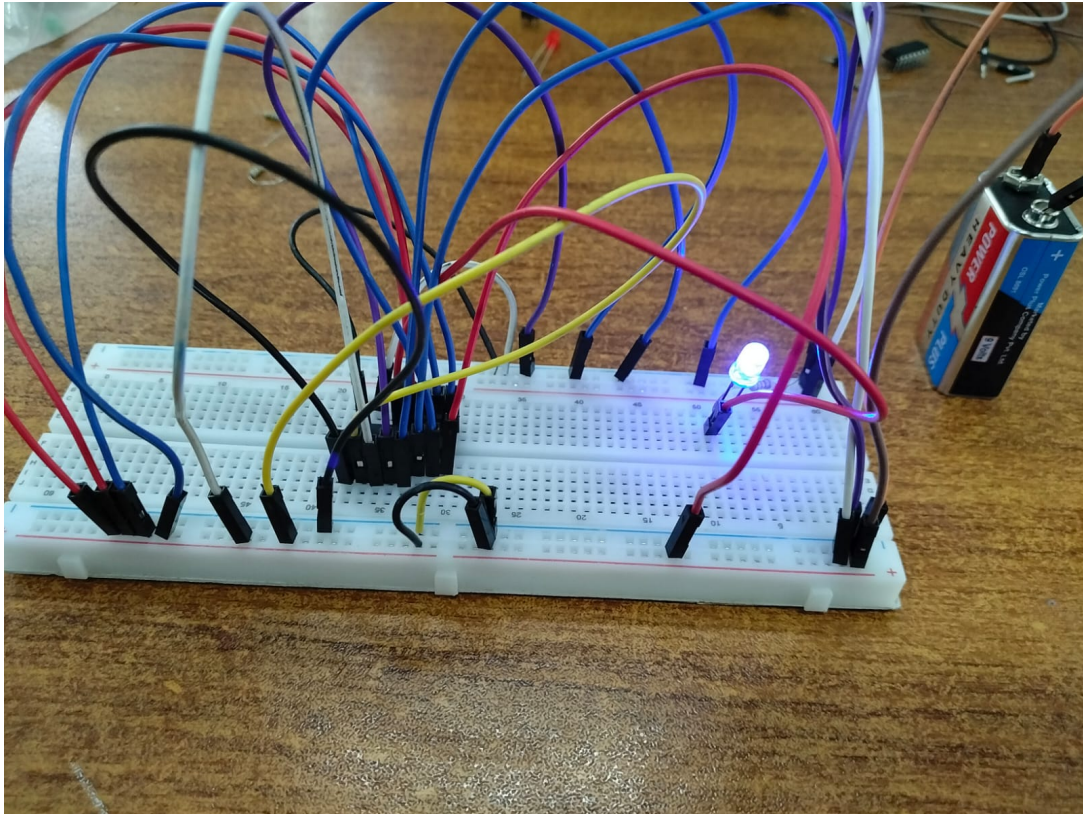


Figure 3: Breadboard implementation of the Multiplexer circuit.

4 Conclusion

The experiment successfully demonstrated the selection and routing capabilities of Digital Multiplexer ICs. By comparing the Electronics Workbench simulations with the hardware implementation, we verified the accuracy of the logic and the functionality of the 74150 and 74151 Integrated Circuits.